Chapter 21

Linear Difference Equations and Z-Transforms

INTRODUCTION

Difference Equations

Difference equations model processes in which we know relationships between changes or differences rather than rates of changes which lead to differential equations. Thus a difference equation is an equation relating various terms of a sequence a_0, a_1, a_2, ... A string loaded with a finite number of beads at equally spaced points leads to a difference equation. Recurrence relations obtained in the solution of DE by power series or Frobenius method are also difference equations. Numerical solution of DE also leads to difference equations.

Z-transforms

Solution of a discrete system, expressed as a difference equation is obtained using z-transform. Discrete analysis played important role in the development of communication engineering.

21.1 LINEAR DIFFERENCE EQUATIONS

Difference equations are functional equations that define sequences just as differential equations define functions. They arise in several situations as follows.

Finance: Compound Interest

Let P_0 be the amount of money deposited (invested) in a bank earning interest periodically say monthly or quarterly or annually. The conversion period r is the time period between interest payment. Let P_n denote the value of the deposit at the end of the nth conversion period, after n interest payments have accrued.

Then in case of simple interest,

$$P_{n+1} = P_n + rP_0$$

which is a first order difference equation

$$P_{n+1} - P_n = rP_0$$

with solution $P_n = P_0(1 + nr)$

In case of compound interest

$$P_{n+1} = P_n + rP_n$$

which is a homogeneous difference equation

$$P_{n+1} - (1 + r)P_n = 0$$

with solution $P_n = P_0(1 + r)^n$. In this discrete process P is a function of an integer n rather than a continuous variable.

Fibonacci Relation:

Suppose there is only one pair of rabbits male and female just born. Further suppose that every month each pair of rabbits that are one month old produce a new pair of offspring of opposite sexes. Then F_n the number of rabbits after n months is given by the recurrence relation.

$$F_n = F_{n-1} + F_{n-2} \quad \text{for} \quad n \geq 2$$

with the initial conditions $F_0 = F_1 = 1$. This gives rise to a second order homogeneous difference equation

$$F_n - F_{n-1} + F_{n-2} = 0$$

By repeated application of the recurrence relation the equation can be solved recursively. Then we get $F_2 = 2, F_3 = 3, F_4 = 5, F_5 = 8, F_6 = 13, F_7 = 21, F_8 = 34$ etc. The disadvantage here is that F_n is calculated upto certain value of n and these values are also dependent on the initial conditions. Instead the general solution

$$F_n = c_1\left(\frac{1+\sqrt{5}}{2}\right)^n + c_2\left(\frac{1-\sqrt{5}}{2}\right)^n$$

is function of n and is independent of the initial conditions.

Differential Equations

In the numerical solution of ordinary differential equations, the derivatives are discretized by replacing them by the finite (forward) differences. This gives rise to difference equations of the higher order. Thus a continuous process described by a differential equation is approximated by a discrete process described by its counterpart a difference equation. (see Chapter on Numerical Analysis 33). For example, in a third order ordinary differential equation

$$a_3 y''' + a_2 y'' + a_1 y' + a_0 y = F(x)$$

the derivatives can be replaced by $y' = \dfrac{y_{n+1} - y_n}{h}$, $y'' = \dfrac{y_{n+2} - 2y_{n+1} + y_n}{h^2}$

$y''' = \dfrac{y_{n+3} - 3y_{n+2} + 3y_{n+1} - y_n}{h^3}$, which results in a third order differences equation of the form

$$b_3 y_{n+3} + b_2 y_{n+2} + b_1 y_{n+1} + b_0 y_n = F(x)$$

Recall that a sequence is a numerical valued function whose domain of definition is the set of integers. It is denoted by $\{a_n\}$ or a_n or $a(n)$.

A kth order linear difference equation in the sequence y_n is of the form

$$a_k y_{n+k} + a_{k-1} y_{n+k-1} + \ldots + a_1 y_{n+1} + \\ + a_0 y_n = f(n) \quad (1)$$

where $n = 0, 1, 2, 3, \ldots$. Thus (1) represents not just a single equation but an infinite system of equations one equation for every n. Here the coefficients $a_0, a_1, \ldots, a_k$ are all constants and do not depend on n. Here $f(n)$ depends only on n. When a_k is chosen as one, (1) is said to be in the standard form. If $f_n \neq 0$ for all n, then (1) is said to be non-homogeneous. Otherwise it is said to be homogeneous if $f_n = 0$ for all n. The *order* of the difference equation (1) is the positive integer k which is the greatest difference in the index of non-zero values of y. Equation (1) is linear because each term in (1) is of first degree (linear) in y_n. Thus (1) is a non-homogeneous kth order linear difference equation with constant coefficients.

Difference equation is also referred to as recurrence relation since it expresses y_{n+k} in terms of one or more of the previous terms (of the sequence) namely $y_{n+k-1}, \ldots, y_{n+1}, y_n$. In this case (1) can be written as $y_{n+k} = -a_{k-1} y_{n+k-1} \ldots - a_1 y_{n+1} - a_0 y_n + f(n)$. The difference equation (1) models a physical system. So f_n is known as system input (system excitation or forcing sequence or driving sequence) while y_n is referred to as system output (system response). The structure of the system is defined by the values of the coefficients and order of the equation. Thus any system output depends on the system input and the structure of the system. The general solution of (1) determines the output y_n which depends only on n (but no longer on the prior terms of the sequence) and describes the complete sequence y_n in the closed form. Thus any sequence y_n that satisfies the difference equation (1) is a solution of (1). The solution of (1) can be obtained by (a) classical approach similar to those used for solving linear non-homogeneous differential equations with constant coefficients (b) Laplace transform method (c) z-transform method (d) recursive method (which is a numerical solution yielding a finite number of terms of the sequence and is disadvantageous because it is influenced by the change of initial conditions). Here we consider only the classical approach since theory of difference equations is analogous to that of differential equations (which is considered in chapter 9).

Homogeneous Equations

First order homogeneous difference equation

Consider a first order linear homogeneous difference equation $y_{n+1} - by_n = 0$ for $n \geq 0$ and b is a constant. Assume the solution of the form $y_n = cr^n$ where $c \neq 0$, $r \neq 0$. Then $y_{n+1} = cr^{n+1}$. Substituting in the given difference equation, we have

$$cr^{n+1} - bcr^n = cr^n(r-b) = 0 \quad \Rightarrow \quad r - b = 0$$

or $b = r$. Thus the general solution of the difference equation $y_{n+1} - by_n = 0$ is given by $y_n = cb^n$. In addition, if a boundary condition $y_0 = d$ then $d = y_0 = cb^0 \quad \therefore c = d$. Then the particular solution is $y_n = db^n$. The solution y_n defines a discrete function whose domain is the set N of all non-negative integers.

Second-order linear homogeneous difference equation with constant coefficients

Consider $a_2 y_{n+2} + a_1 y_{n+1} + a_0 y_n = 0$ (2)

Assume $y_n = cr^n$ (with $c \neq 0, r \neq 0$) (3)

as a solution of (2). Then substituting (3) in (2), we get

$$a_2 cr^{n+2} + a_1 cr^{n+1} + a_0 cr^n = 0$$

or
$$cr^n(a_2 r^2 + a_1 r + a_0) = 0$$

Thus (3) is solution of (2) if

$$a_2 r^2 + a_1 r + a_0 = 0 \tag{4}$$

since $c \neq 0$ and $r \neq 0$. The equation (4) which is a quadratic in r is known as the *characteristic or auxiliary equation* of (2). Three cases arise.

Case 1: When the roots of the characteristic equation are *real and distinct* given by r_1 and r_2 then r_1^n and r_2^n are two linearly independent solutions. Thus the general solution of (2) is

$$\boxed{y_n = c_1 r_1^n + c_2 r_2^n} \tag{5}$$

where c_1, c_2 are two arbitrary constants.

Case 2: If the roots are *real and equal*, say r, then the general solution of (2) is

$$\boxed{y_n = c_1 r^n + c_2 \cdot n \cdot r^n = (c_1 + n \cdot c_2) r^n} \tag{6}$$

Case 3: Suppose the roots of (4) are *complex conjugate* given by $a \pm bi$. Then the general solution of (2) is

$$\boxed{y_n = r^n(c_1 \cos n\theta + c_2 \sin n\theta)} \tag{7}$$

where $r = \sqrt{a^2 + b^2}$, $\tan\theta = \frac{b}{a}$.

This analysis can easily be extended to kth order difference equation by considering the nature of the k roots of the auxilary equation which is a kth degree polynomial. (see Chapter 9).

Note 1: The forward-difference or advancing-difference operator Δ is defined by $\Delta f_k = f_{k+1} - f_k$

Note 2: The shift operator E is defined as the operator that increases the argument of a function by one tabular interval. Thus

$$Ef_k = Ef(x_k) = f(x_k + h) = f(x_{k+1}) = f_{k+1}$$

Note 3: Δ and E are related by $E = 1 + \Delta$.

Note 4: The difference equation

$$a_k y_{n+k} + a_{k-1} y_{n+k-1} + \ldots + a_1 y_{n+1} + a_0 y_n = f(n) \tag{1}$$

can be written in terms of E as follows

$$(a_k E^k + a_{k-1} E^{k-1} + \ldots + a_1 E + a_0) y_n = f(n) \tag{8}$$

Non-homogeneous Equations

The general solution of a non-homogeneous linear difference equation with constant coefficients (1) is the sum of the complementary function and any particular solution. Here the complementary function (C.F.) of (1) is the general solution of the corresponding homogeneous equation (2). Particular solution, more often known as particular integral of (1) can be obtained by (a) method of undetermined coefficients (b) short cut inverse operator methods.

(a) In the method of undetermined coefficients

The particular integral is assumed in a particular form depending on the form of the RHS function f_n.

(b) Inverse operator methods

The non-homogeneous equation (8) can be written as

$$F(E) y_n = f(n) \tag{9}$$

where $F(E) = (a_k E^k + a_{k-1}E^{k-1} + \ldots + a_1 E + a_0)$ (10)
is a function of the operator E.

Then the particular integral is

$$\text{P.I.} = \frac{1}{F(E)} f(n)$$

Case 1: If $f(n) = a^n$ then

$\text{P.I.} = \frac{1}{F(E)} a^n = \frac{1}{F(a)} a^n$, provided $F(a) \neq 0$.

Case 2: Failure case: If $F(a) = 0$, then

$$\text{P.I.} = \frac{1}{(E-a)^3} a^n = \frac{n(n-1)(n-2)}{3!} a^{n-3}$$

Case 3: If $f(n) = \sin \alpha n$ then

$$\text{P.I.} = \frac{1}{F(E)} \sin \alpha n = \frac{1}{F(E)} \left(\frac{e^{i\alpha n} - e^{-i\alpha n}}{2i} \right)$$

$$= \frac{1}{2i} \left[\frac{1}{F(E)} a^n - \frac{1}{F(E)} b^n \right]$$

where $a = e^{i\alpha n}$ and $b = e^{-i\alpha n}$.

Similarly if $f(n) = \cos \alpha n$, replace

$$\cos \alpha_n = \frac{1}{2}(e^{i\alpha n} + e^{-i\alpha n}) = \frac{1}{2}(a^n + b^n)$$

Case 4: If $f(n) = n^m$ or polynomial in n. Replace E by $1 + \Delta$ and expand $1/F(1+\Delta)$ in binomial series in ascending powers of Δ upto Δ^m. Express $f(n)$ in factorials and use $\Delta[x]^n = n[x]^{n-1}$

Case 5: If $f(n) = a^n V(n)$ where $V(n)$ is a polynomial in n. Then

$$\text{P.I.} = \frac{1}{F(E)} \{a^n V(n)\} = a^n \frac{1}{F(aE)} V(n)$$

A. Homogeneous difference equations

Worked Out Examples

First-order difference equation

Example 1: Find the general solution of the first order difference equation $2a_n - 3a_{n-1} = 0$, $n \geq 1$, $a_4 = 81$.

Solution: The general solution of $a_n - \frac{3}{2}a_{n-1} = 0$ or $a_{n+1} - \frac{3}{2}a_n = 0$ for $n \geq 0$ is $a_n = Ad^n = A\left(\frac{3}{2}\right)^n$.

Since $81 = a_4 = A\left(\frac{3}{2}\right)^4$

$\therefore$ $A = 16$. Thus the unique solution is $a_n = 16\left(\frac{3}{2}\right)^n$ for $n \geq 0$.

Finance: Compound interest

Example 2: If Raju invests Rs 1000 at 6% interest compounded quarterly, how many month must he wait for his money to double (Note that Raju can *not* withdraw the money before the quarter is up). How many months it trebles.

Solution: Annual interest rate is 6% so the quarterly rate is $\frac{6\%}{4} = \frac{3}{2}\% = \frac{3}{200} = 0.015$. For $0 \leq n \leq 4$, P_n denotes the value of Raju's deposit at the end of n quarters.

Then $P_{n+1} = P_n + 0.015P_n$ where $0.015P_n$ is the interest earned on P_n during $(n+1)$th quarter. Here $P_0 = 1000$. The solution of the difference equation $P_{n+1} - 1.015P_n = 0$ is $P_n = P_0(1.015)^n = 1000(1.015)^n$. If the money doubles then $P_n = 2P_0 = 2000$. Then $2000 = 1000(1.015)^n$ or $\frac{\ln 2}{\ln(1.015)} = n$ or $n = 46.56 \approx 47$ quarters. So money doubles in $47 \times 3 = 141$ months.

If money trebles then $3000 = 1000(1.015)^n$ so

$$n = \frac{\ln 3}{\ln(1.015)} = 73.80 \approx 74,$$

i.e. in $74 \times 3 = 222$. Thus money trebles in 222 months.

Formation of difference equation

Example 3: Form the differences equation corresponding to the family of curves by eliminating the two arbitrary constants.

(a) $y_n = A3^n + B5^n$ (b) $y(x) = ax + b2^x$

Solution: (a) From $y_n = A3^n + B5^n$ with $n = n+1$ and $n+2$ we get $y_{n+1} = A3^{n+1} + B5^{n+1}$ $= 3A3^n + 5B5^n$ and $y_{n+2} = A3^{n+2} + B5^{n+2} =$

$9A3^n + 25B5^n$. Eliminating A and B, we get

$$\begin{vmatrix} y_n & 1 & 1 \\ y_{n+1} & 3 & 5 \\ y_{n+2} & 9 & 25 \end{vmatrix} = 0.$$

Expanding the determinant

$$y_n(75-45) - 1(25y_{n+1} - 5y_{n+2}) + 1(9y_{n+1} \\ -3y_{n+2}) = 0$$

or the required differences equation is

$$30y_n - 16y_{n+1} - 2y_{n+2} = 0$$

(b) Rewriting $y_n = a_n + b2^n$, we get

$$\Delta y_n = y_{n+1} - y_n = [a(n+1) + b2^{n+1}] \\ -[an + b2^n]$$

$$\Delta y_n = a + b2^n(2-1) = a + b2^n$$

Now $\Delta^2 y_n = \Delta y_{n+1} - \Delta y_n$

$$= (a + b2^{n+1}) - (a + b2^n)$$

$$= b2^n(2-1) = b2^n$$

Thus $b = \frac{\Delta^2 y_n}{2^n}$.

Substituting b, we get

$$\Delta y_n = a + b2^n = a + \Delta^2 y_n$$

or $\quad a = \Delta y_n - \Delta^2 y_n$

Now eliminating a and b

$$y_n = a_n + b2^n = (\Delta y_n - \Delta^2 y_n)n + \Delta^2 y_n$$

$$y_n = \Delta y_n + (1-n)\Delta^2 y_n$$

since $\Delta^2 y_n = \Delta y_{n+1} - \Delta y_n =$

$$(y_{n+2} - y_{n+1}) - (y_{n+1} + y_n)$$

$$= y_{n+2} - 2y_{n+1} + y_n$$

We get

$$y_n = (y_{n+1} - y_n) + (1-n)(y_{n+2} - 2y_{n+1} + y_n)$$

or $\quad (1-n)y_{n+2} + y_{n+1}(1 - 2 + 2n)$

$$+y_n(-2 + 1 - n)$$

Thus the difference equation is

$$(1-x)y_{x+2} + (2x-1)y_{x+1} - (x+1)y_x = 0$$

Real, distinct roots

Example 4: Solve $y_{n+2} - 3y_{n+1} - 10y_n = 0$

Solution: The auxiliary equation (A.E.) is obtained by assuming $y_n = cr^n$ and substituting in the given difference equation. Thus $y_{n+1} = cr^{n+1}$, $y_{n+2} = cr^{n+2}$ so

$$cr^{n+2} - 3cr^{n+1} - 10cr^n = 0$$

$$\text{or} \quad r^2 - 3r - 10 = 0$$

is the characteristic equation with distinct real roots $r_1 = 5$ and $r_2 = -2$ (since $r^2 - 3r - 10 = (r-5)(r+2) = 0$). Symbolically the A.E. is obtained by replacing y_n by 1, y_{n+1} by r, y_{n+2} by r^2 (or equivalently E by r and E^2 by r^2).

Then $y_n = 5^n$ and $y_n = (-2)^n$ are two linearly independent solutions because one is *not* a multiple of the other. Thus the general solution of the given difference solution is

$$y_n = c_1 5^n + c_2(-2)^n$$

Equal roots

Example 5: Solve $a_n - 6a_{n-1} + 9a_{n-2} = 0$, $n \geq 2$, $a_0 = 5$, $a_1 = 12$.

Solution: The auxilary equation is $r^2 - 6r + 9 = 0$ or $(r-3)^2$. So the roots are $r = 3, 3$, real equal. The general solution is

$$a_n = c_1 3^n + c_2 \cdot n3^n$$

Since $5 = a_0 = c_1 \cdot 1 + c_2 \cdot 0 \cdot 1 \qquad \therefore \quad c_1 = 5$

Since $12 = a_1 = 5 \cdot 3 + c_2 \cdot 1 \cdot 3 \qquad \therefore \quad c_2 = -1$

The required (particular) solution satisfying the given two conditions is

$$a_n = 53^n - n3^n = (5-n)3^n$$

Complex conjugate roots

Example 6: Solve $(E^3 - E^2 + E - 1)y = 0$ with $y_0 = 1$, $y_1 = 0$ and $y_2 = 2$.

Solution: Here E is the *shift* operator. So the given equation is $y_{n+3} - y_{n+2} + y_{n-1} - y_n = 0$.

Its auxiliary equation is $m^3 - m^2 + m - 1 = 0$ or $(m-1)(m^2+1) = 0$. So $m = 1$ and $m = \pm i$ are the

three roots. For the complex conjugate roots $a = 0$, $b = 1$, so in the polar form $i = 1 \cdot e^{i\frac{\pi}{2}}$. Here $r = \sqrt{a^2+b^2} = 1$, $\theta = \tan\frac{b}{a} = \infty \quad \therefore \quad \theta = \pi/2$.
Then the general solution is

$$y_n = c_1 1^n + r^n(c_2 \cos n\theta + c_3 \sin n\theta)$$

$$y_n = c_1 + 1^n \left(c_2 \cos n\frac{\pi}{2} + c_3 \sin n\frac{\pi}{2}\right)$$

Since $1 = y_0 = c_1 + c_2 + c_3 \cdot 0 \quad \therefore c_1 + c_2 = 1$
Since $0 = y_1 = c_1 + c_2 \cdot 0 + c_3 \quad \therefore c_1 + c_3 = 0$
Since $2 = y_2 = c_1 - c_2 + c_3 \cdot 0 \quad \therefore c_1 - c_2 = 2$
Solving $c_1 = \frac{3}{2}$, $c_2 = -\frac{1}{2}$, $c_3 = -\frac{3}{2}$.
Thus the particular solution is

$$y_n = \frac{3}{2} - \frac{1}{2}\cos\frac{n\pi}{2} - \frac{3}{2}\sin\frac{n\pi}{2}$$

Simultaneous equations

Example 7: Solve $x_{n+1} - 7x_n - 10y_n = 0$ and $y_{n+1} - x_n - 4y_n = 0$, with $x_0 = 3$ and $y_0 = 2$.

Solution: Introducing the shift operator E notation, the given two equations can be written as

$$(E-7)x_n - 10y_n = 0 \tag{1}$$

$$-x_n + (E-4)y_n = 0 \tag{2}$$

Multiplying (1) by (E-4) and (2) by 10 and adding, we get

$$(E-7)(E-4)x_n - 10x_n = 0$$

or $$(E^2 - 11E + 28 - 10)x_n$$

$$= (E^2 - 11E + 18)x_n = 0.$$

Its auxiliary equation is $r^2 - 11r + 18 = 0$ with two real distinct roots 2 and 9. Then

$$x_n = c_1 2^n + c_2 9^n \tag{3}$$

Using $3 = x_0 = c_1 \cdot 1 + c_2 \cdot 1$, we get $c_1 + c_2 = 3$ (4)
Substituting (3) in (1) we have

$$(c_1 2^{n+1} + c_2 9^{n+1}) - 7(c_1 2^n + c_2 9^n) - 10y_n = 0$$

So $y_n = \frac{1}{10}[-5c_1 2^n + 2c_2 9^n]$

Using $2 = y_0 = \frac{1}{10}[-5 \cdot c_1 \cdot 1 + 2c_2 \cdot 1]$, we get

$$-5c_1 + 2c_2 = 20 \tag{5}$$

Solving (4) and (5) we have $c_1 = -2$, $c_2 = 5$. Thus the required solution is

$$x_n = -2 \cdot 2^n + 5 \cdot 9^n$$

and $y_n = \frac{1}{10}[102^n + 109^n] = 2^n + 9^n$

Determinant

Example 8: Express the following nth order determinant D_n as an explicit function of n. When $a > 2$, show that $D_n = \frac{\sinh(n+1)\mu}{\sinh\mu}$ where $\cos h\mu = \frac{a}{2}$. What is the value of D_n when (i) $a = 2$ (ii) $a = -2$ where

$$D_n = \begin{vmatrix} a & 1 & 0 & . & . & . & 0 & 0 \\ 1 & a & 1 & . & . & . & 0 & 0 \\ 0 & 1 & a & . & . & . & 0 & 0 \\ . & . & . & . & . & . & . & . \\ 0 & 0 & 0 & . & . & . & a & 1 \\ 0 & 0 & 0 & . & . & . & 1 & a \end{vmatrix}$$

Solution: Expanding D_n in terms of the elements in the first row (or column), we get

$$D_n = a \cdot \begin{vmatrix} a & 1 & 0 & \dots & 0 & 0 \\ 1 & a & 1 & \dots & 0 & 0 \\ 0 & 1 & a & \dots & \dots & \dots \\ \dots & \dots & 0 & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & a & 1 \\ 0 & 0 & 0 & \dots & 1 & a \end{vmatrix} -$$

$$-1 \cdot \begin{vmatrix} 1 & 1 & \dots & \dots & 0 & 0 \\ 0 & a & 1 & \dots & 0 & 0 \\ 0 & 1 & a & 1\dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & a & 1 \\ 0 & 0 & 0 & \dots & 1 & a \end{vmatrix}$$

The first determinant in RHS is identical in structure to D_n except that it contains only $n-1$ rows and $n-1$ columns. Thus the first determinant in RHS is D_{n-1}. The second determinant in RHS does not have the form of D_n. But expanding the second determinant by the first row, we get

$$D_n = a \cdot D_{n-1} - 1 \cdot 1 \begin{vmatrix} a & 1 & 0 & \ldots & 0 & 0 \\ 1 & a & 1 & \ldots & 0 & 0 \\ 0 & 1 & a & \ldots & 0 & 0 \\ \ldots & \ldots & \ldots & \ldots & 0 & 0 \\ 0 & 0 & 0 & \ldots & a & 1 \\ 0 & 0 & 0 & \ldots & 1 & a \end{vmatrix}$$

Now the second determinant in RHS has the same structure as D_n but is of $(n-2)$ order. Thus

$$D_n = aD_{n-1} - D_{n-2}$$

$$\text{or} \quad D_{n+2} - aD_{n+1} + D_n = 0$$

i.e., $$(E^2 - aE + 1)D_n = 0 \tag{1}$$

The auxiliary equation is

$$m^2 - am + 1 = 0$$

Its roots are given by

$$m = \frac{a \pm \sqrt{a^2-4}}{2} = \frac{a \pm b}{2} \quad \text{where } b = \sqrt{a^2-4} \tag{2}$$

If $a > 2$ then $b > 0$ so the roots are real, distinct given by

$$m_1 = \frac{a+b}{2}, \; m_2 = \frac{a-b}{2} \tag{3}$$

Then the general solution of the difference equation (1) is

$$D_n = c_1 m_1^n + c_2 m_2^n$$

$$D_n = c_1 \left(\frac{a+b}{2}\right)^n + c_2 \left(\frac{a-b}{2}\right)^n \tag{4}$$

Thus (4) expresses the nth order determinant D_n as an explicit function of n (in terms of the parameter 'a').

Put $\cosh\mu = \frac{a}{2}$ then $b = a^2 - 4 = \sqrt{4\cosh^2\mu - 4}$ so 'b' $= 2\sinh\mu$. Then $m_1 = \frac{a+b}{2} = \frac{2\cdot\cosh\mu + 2\cdot\sinh\mu}{2} = e^{\mu}$

Similarly $m_2 = \frac{a-b}{2} = \cosh\mu - \sinh\mu = e^{-\mu}$, then the general solution (4) takes the form

$$D_n = c_1(e^{\mu})^n + c_2(e^{-\mu})^n$$

$$= c_1(\cosh\mu n + \sinh\mu n) + c_2(\cosh\mu n - \sinh\mu n)$$

$$D_n = A\cosh\mu n + B\sinh\mu n$$

where $A = c_1 + c_2$ and $B = c_1 - c_2$. Choosing $A = 1$ and $B = \coth\mu$, we get

$$D_n = \cosh\mu n + \coth\mu \cdot \sinh\mu n = \frac{\sinh(n+1)\mu}{\sinh\mu}$$

Now for $n = 1$, $D_1 = a$ and $n = 2$, $D_2 = \begin{vmatrix} a & 1 \\ 1 & a \end{vmatrix} = a^2 - 1$. Thus $D_1 = a = 2\cosh\mu$, $D_2 = 4\cosh^2\mu - 1 = 2\cosh 2\mu$. Using $2\cosh\mu = a = D_1 = A\cosh\mu + B\sinh\mu$. So $A = 2$ and $B = 0$. Using $2\cosh 2\mu = D_2 = 2\cdot\cosh 2\mu + 0\cdot\sinh\cdot 2\mu = 2\cosh 2\mu$ which is true.

Case 1: For $a = 2$, $b = 0$, the roots are real and equal given by $m_1 = m_2 = 1$. Then the general solution is

$$D_n = (c_1 + nc_2)\cdot 1^n = c_1 + nc_2$$

For $n = 1$, $D_1 = a = 2 = c_1 + 1\cdot c_2$.
For $n = 2$, $D_2 = a^2 - 1 = 4 - 1 = 3 = c_1 + 2c_2$
Solving $c_1 = c_2 = 1$. Thus when $a = 2$, $D_n = 1 + n$.

Case 2: When $a = -2$, the general solution is $D_n = (c_1 + c_2\cdot n)(-1)^n$
For $n = 1$, $D_1 = a = -2 = (c_1 + c_2)(-1)$
For $n = 2$, $D_2 = a^2 - 1 = 3 = c_1 + 2c_2$
Solving $c_1 = c_2 = 1$. Thus when $a = -2$, we get

$$D_n = (1+n)(-1)^n$$

Example 9: A seed of a particular plant produces 8-fold when one year old and produces 18-fold when two or more years old. If a_n denotes the number of seeds produced at the end of the nth year, express a_n in term of n.

Solution: At the end of $(n+1)$th year, the a_n seeds which are one year old produces 8-fold. The a_{n-1} seeds which are two years old produces 18-fold. Similarly the $a_{n-2}, a_{n-3}, \ldots a_2, a_1$ which are more than two years old also produces 18 fold. Thus the recurrence relation is

$$a_{n+1} = 8(a_n) + 18(a_{n-1} + a_{n-2} + \ldots + a_2 + a_1)$$

Similarly

$$a_{n+2} = 8(a_{n+1}) + 18(a_n + a_{n-1} + \ldots + a_3 + a_2 + a_1)$$

Subtracting

$$a_{n+2} - a_{n+1} = 8a_{n+1} + 18a_n - 8a_n$$

or the difference equation is

$$a_{n+2} - 9a_{n+1} - 10a_n = 0.$$

Its characteristic equation is

$$m^2 - 9m - 10 = (m - 10)(m + 1) = 0$$

with real distinct roots $m_1 = 10, m_2 = -1$. The general solution is

$$a_n = c_1(10)^n + c_2(-1)^n$$

Non-homogeneous difference equations

WORKED OUT EXAMPLES

Method of undetermined coefficients

Example 1: Determine a formula for the sum of the first n cubes of natural numbers.

Solution: Let S_n be the partial sum of the cubes of the first n natural numbers. Then

$$S_n = 1^3 + 2^3 + 3^3 + \ldots + n^3 = \sum_{i=1}^{n} i^3$$

Now $S_{n+1} = 1^3 + 2^3 + 3^3 + \ldots + n^3 + (n+1)^3$
$= \sum_{i=1}^{n+1} i^3$

Subtracting

$$S_{n+1} - S_n = (n+1)^3 \qquad (1)$$

This is a first order non-homogeneous difference equation with auxiliary equation $m - 1 = 0$. So the complementary function is

$$S_c = c \cdot 1^n = c \qquad (2)$$

where c is an arbitrary constant. To determine the particular integral, use the method of undetermined coefficients. Since the RHS is a function of the type

$$(n+1)^3 = n^3 + 3n^2 + 3n + 1$$

We assume the particular integral of the form

$$S_n = An^4 + Bn^3 + Cn^2 + Dn \qquad (3)$$

(Even if we take an additional constant E, it will become redundant). Substituting (3) in (1), we get

$$S_{n+1} - S_n = [A(n+1)^4 + B(n+1)^3 + C(n+1)^2 + D(n+1)] - [An^4 + Bn^3 + Cn^2 + Dn] = (n+1)^3 \qquad (5)$$

The unknown coefficients A, B, C, D will be determined by equating the coefficients of like powers of n on both sides of (5).

$$A(n^4+4n^3+6n^2+4n+1)+B(n^3+3n^2+3n+1)$$
$$+C(n^2 + 2n + 1) - (An^4 + Bn^3 + Cn^2 + Dn)$$
$$= n^3 + 3n^2 + 3n + 1$$

or $(4A)n^3 + (6A + 3B)n^2 + (4A + 3B + 2C)n$

$$+(A + B + C + D) = n^3 + 3n^2 + 3n + 1$$

Then $4A = 1,\ 6A + 3B = 3,\ 4A + 3B + 2C = 3,$ and $A + B + C + D = 1$
Solving $A = \frac{1}{4}, B = \frac{1}{2}, C = \frac{1}{4}, D = 0$. Thus the particular integral is $\frac{1}{4}n^4 + \frac{1}{2}n^3 + \frac{1}{4}n^2$. Then the general solution of the difference equation (1) is given by

$$S_n = \text{C.F.} + \text{P.I.} = c + \frac{1}{4}n^4 + \frac{1}{2}n^3 + \frac{1}{4}n^2 \qquad (6)$$

For $n = 1$, we know that $S_1 = 1^3 = 1$. Using this in (6), we get

$$1 = S_1 = c + \frac{1}{4} + \frac{1}{2} + \frac{1}{4} \qquad \therefore\ c = 0$$

Thus the formula expressing the sum of cubes of the first n natural numbers is

$$S_n = \sum_{i=1}^{n} i^3 = 0 + \frac{1}{4}n^4 + \frac{1}{2}n^3 + \frac{1}{4}n^2$$

$$S_n = \frac{n^2(n+1)^2}{4}$$

Short cut inverse operator methods

Power function

Example 2: Solve $a_{n+2} - 6a_{n+1} + 5a_n = 2^n$ with $a_0 = 0, a_1 = 0$.

Solution: The auxiliary equation is $m^2 - 6m + 5 = 0$ with real distinct roots $m_1 = 1, m_2 = 5$. So the complementary function is $c_1 \cdot 1^n + c_2 5^n$. The particular integral is obtained by inverse operator method. The given difference equation is rewritten in terms of the shift operator E as $(E^2-6E+5)a_n = 2^n$. Then the particular integral is

$$\text{P.I.} = \frac{1}{E^2-6E+5}\cdot 2^n = \frac{1}{2^2-6\cdot 2+5}\cdot 2^n$$
$$= -\frac{1}{3}2^n$$

Here E is replaces by 2. Thus the general solution is

$$a_n = \text{C.F.} + \text{P.I.} = c_1 + c_2 5^n - \frac{1}{3}2^n$$

Since $0 = a_0 = c_1 + c_2 - \frac{1}{3}$ $\therefore$ $c_1 + c_2 = \frac{1}{3}$
Since $0 = a_1 = c_1 + 5c_2 - \frac{2}{3}$ $\therefore$ $c_1 + 5c_2 = \frac{2}{3}$
Solving $c_1 = \frac{1}{4}$, $c_2 = \frac{1}{12}$. Then

$$a_n = \frac{1}{4} + \frac{1}{12}5^n - \frac{1}{3}2^n$$

Failure case

Example 3: Solve $y_{n+3} - 12y_{n+2} + 48y_{n+1} - 64y_n = 5\cdot 4^n$.

Solution: A.E. is $m^3 - 12m^2 + 48m - 64 = 0$ or $(m-4)^3 = 0$. The roots are real, equal repeated thrice $m = 4, 4, 4$. So the

$$\text{C.F.} = c_1 4^n + c_2\cdot n\cdot 4^n + c_3 n^2 4^n$$

$$\text{Now P.I.} = \frac{1}{E^3-12E^2+48E-64}5\cdot 4^n$$
$$= \frac{5}{(E-4)^3}\cdot 4^n$$

Here E can *not* be replaced by 4. Then by result (case 2 on page 21.4) we have

$$\text{P.I.} = 5\cdot\frac{n(n-1)(n-2)}{3!}4^{n-3}$$

Thus the general solution is

$$y_n = \text{C.F.} + \text{P.I.} = (c_1 + c_2\cdot n + c_3\cdot n^2)4^n +$$
$$+\frac{5n(n-1)(n-2)}{3!}4^{n-3}$$

Trigonometric function

Example 4: Find the general solution of $(E^2+4)y_n = \cos\alpha n$.

Solution: A.E. is $m^2+4=0$ with conjugate complex roots $\pm 2i$. So $a=0$, $b=2$, then $r=\sqrt{a^2+b^2}=2$, and $\tan\theta = \frac{2}{0} = \infty$. $\therefore$ $\theta = \frac{\pi}{2}$. Thus

$$\text{C.F.} = r^n(c_1\cos n\theta + c_2\sin n\theta)$$
$$\text{C.F.} = 2^n\left(c_1\cos\frac{n\pi}{2} + c_2\sin\frac{n\pi}{2}\right)$$

Now P.I. $= \frac{1}{E^2+4}\cos\alpha n = \frac{1}{(E+2i)(E-2i)}\cos\alpha n$.
But $\cos\alpha n = \frac{1}{2}(e^{in\alpha} + e^{-in\alpha})$. So

$$\text{P.I.} = \frac{1}{(E+2i)(E-2i)}\left\{\frac{1}{2}(e^{in\alpha}+e^{-in\alpha})\right\}$$
$$= \frac{1}{(E+2i)(E-2i)}\frac{1}{2}(a^n+b^n)$$

where $a = e^{i\alpha}$, $b = e^{-i\alpha}$

$$= \frac{1}{2}\frac{1}{(a+2i)(a-2i)}a^n + \frac{1}{2}\frac{1}{(b+2i)(b-2i)}b^n$$
$$= \frac{1}{2}\cdot\frac{1}{a^2+4}a^n + \frac{1}{2}\frac{1}{(b^2+4)}b^n$$
$$= \frac{1}{2}\left[\frac{1}{e^{2i\alpha}+4}e^{i\alpha n} + \frac{1}{e^{-2i\alpha}+4}e^{-i\alpha n}\right]$$
$$= \frac{1}{2}\left[\frac{(e^{-2i\alpha}+4)e^{i\alpha n} + (e^{2i\alpha}+4)e^{-i\alpha n}}{(e^{2i\alpha}+4)(e^{-2i\alpha}+4)}\right]$$
$$= \frac{4\cos\alpha n + \frac{1}{2}\{e^{i\alpha(n-2)} + e^{i\alpha(n-2)}\}}{1+4(e^{2i\alpha}+e^{-2i\alpha})+16}$$
$$= \frac{4\cos\alpha n + \cos\alpha(n-2)}{17+8\cos 2\alpha}$$

Thus the general solution is

$$y_n = 2^n\left(c_1\cos\frac{n\pi}{2} + c_2\sin\frac{\pi}{2}\right) +$$
$$+\left(\frac{4\cos\alpha n + \cos\alpha(n-2)}{17+8\cos 2\alpha}\right)$$

Polynomial

Example 5: Solve $a_{n+2} - 5a_{n+1} + 6a_n = 2n^2 - 6n - 1$.

Solution: A.E. is $m^2 - 5m + 6 = 0$ with real distinct roots 2 and 3. So C.F. $= c_1 2^n + c_2 3^n$. The given difference equation in shift oeprator E is

$$(E^2 - 5E + 6)a_n = 2n^2 - 6n - 1$$

So P.I. $= \dfrac{1}{E^2 - 5E + 6)}(2n^2 - 6n - 1)$

Replace E by $1 + \Delta$ then

$$E^2 - 5E + 6 = (1 + \Delta)^2 - 5(1 + \Delta) + 6$$

$$= 1 + \Delta^2 + 2\Delta - 5 - 5\Delta + 6 = \Delta^2 - 3\Delta + 2$$

Then

$$\text{P.I.} = \frac{1}{\Delta^2 - 3\Delta + 2}(2n^2 - 6n - 1)$$

$$= \frac{1}{2}\frac{1}{1 + \left(\frac{\Delta^2 - 3\Delta}{2}\right)}(2n^2 - 6n - 1)$$

Expanding in binomial series

$$= \frac{1}{2}\left[1 - \left(\frac{\Delta^2 - 3\Delta}{2}\right) + \left(\frac{\Delta^2 - 3\Delta}{2}\right)^2 + \ldots\right] \times$$

$$\times (2n^2 - 6n - 1)$$

Neglecting powers of Δ more than 2, we get

$$\text{P.I.} = \frac{1}{2}\left[1 + \frac{3}{2}\Delta + \frac{7}{4}\Delta^2\right][2n^2 - 2n - 4n - 1]$$

$$= \frac{1}{2}\left[1 + \frac{3}{2}\Delta + \frac{7}{4}\Delta^2\right]\{2[n]^2 - 4[n] - 1\}$$

where $[n]^2 = n(n-1)$. Then

$$\text{P.I.} = \frac{1}{2}\left[\{2[n]^2 - 4[n] - 1\} + \frac{3}{2}\{4[n] - 4\} + \frac{7}{4}\{4\}\right]$$

$$= \frac{1}{2}\left[\{2n^2 - 6n - 1\} + 6\{(n-1)\} + 7\right] = n^2$$

Therefore the general solution is

$$a_n = c_1 2^n + c_2 3^n + n^2$$

Exponential shift

Example 6: Solve $(E^2 + E - 56)a_n = 2^n(n^2 - 3)$

Solution: A.E. is $m^2 + m - 56 = (m + 8)(m - 7) = 0$ with distinct real roots -8, 7. So C.F. $= c_1(-8)^n + c_2(7)^n$.

Since the R.H.S. is of the form $a^n F(n)$, apply shift result for obtaining particular integral. So replace E by $2E$, we get

$$\text{P.I.} = \frac{1}{E^2 + E - 56}\{2^n(n^2 - 3)\}$$

$$= \frac{2^n}{(2E)^2 + 2E - 56}\{(n^2 - 3)\}.$$

$$= \frac{2^n}{2(2E^2 + E - 28)}(n^2 - 3)$$

$$= \frac{2^n}{2[2(1 + \Delta)^2 + (1 + \Delta) - 28)]}(n^2 - 3)$$

where E is replaced by $1 + \Delta$.

$$\text{P.I.} = \frac{2^n}{2[2\Delta^2 + 5\Delta - 25]}(n^2 - 3)$$

$$= -\frac{2^n}{50}\left[1 - \left(\frac{5\Delta + 2\Delta^2}{25}\right)\right]^{-1}(n^2 - 3)$$

Expanding in binomial series we get

$$\text{P.I.} = -\frac{2^n}{50}\left[1 + \frac{5\Delta + 2\Delta^2}{25} + \left(\frac{5\Delta + 2\Delta^2}{25}\right)^2 + \ldots\right] \times$$

$$\times (n^2 - 3).$$

Neglecting powers of Δ more than 2 we have

$$\text{P.I.} = -\frac{2^n}{50}\left[1 + \frac{5}{25}\Delta + \frac{3}{25}\Delta^2\right]\{[n]^2 + [n] - 3\}$$

where $n^2 - 3 = n^2 - n + n - 3 = n(n-1) + n - 3 = [n]^2 + [n] - 3$.

$$\text{P.I.} = -\frac{2^n}{50}[\{[n]^2 + [n] - 3\} + \frac{5}{25}\{2[n] + 1\} + \frac{3}{25}\{2\}]$$

$$\text{P.I.} = -\frac{2^n}{50}\left[n^2 + \frac{2}{5}n - \frac{64}{25}\right]$$

Therefore the general solution is

$$a_n = c_1(-8)^n + c_2(7)^n - \frac{2^{n-1}}{25}\left(n^2 + \frac{2}{5}n - \frac{64}{25}\right)$$

Homogeneous difference equations

EXERCISE

First order

Solve the difference equation

1. $a_n - 7a_{n-1} = 0$ when $n \geq 1$ and $a_2 = 98$

Ans. $a_n = 2(7)^n$ for $n \geq 0$

2. $a_{n+1}^2 - 5a_n^2 = 0$ when $a_n > 0$ for $n \geq 0$ and $a_0 = 2$. Also find a_{12}.

Ans. $a_n = 2(\sqrt{5})^n$, $a_{12} = 2(\sqrt{5})^{12} = 31,250$

Hint: $a_n^2 = b_n$ and solve for b_n.

3. How much will be the deposit of Rs 1000 a year later, if the bank pays 6% annual interest, compounding the interest monthly.

Ans. Rs 1061.68

Hint: $P_{n+1} = P_n + 0.005P_n$, $P_n = P_0(1.005)^n$, $P_0 = 1000$, $n = 12$

4. How much will be Rs 10,000 after 30 years in a bank yielding 11% per year interest compounded annually.

Ans. Rs 228,922.97

Hint: $P_{n+1} = P_n + 0.11P_n$, $P_n = P_0(1.11)^n$, $P_0 = 10{,}000$, $n = 30$

5. Solve the following difference equations
(a) $a_n - 3a_{n-1} = 0$, $a_0 = 2$

Ans. $a_n = 2 \cdot 3^n$

(b) $a_n - 2a_{n-1} + 1 = 0$, $a_0 = 1$

Ans. $a_n = 1$

(c) $a_n - 2na_{n-1} = 0$, $a_0 = 1$,

Ans. $a_n = 2^n \cdot n!$

Second and higher order

Solve the following difference equations

6. $y_{n+2} + y_{n+1} - 6y_n = 0$ with $a_0 = -1$, $a_1 = 8$

Ans. $y_n = 2^n - 2(-3)^n$

7. $F_{n+2} - F_{n+1} - F_n = 0$ for $n \geq 0$, with $F_0 = 0$, $F_1 = 1$.

Ans. $F_n = \frac{1}{a}\left[\left(\frac{1+a}{2}\right)^n - \left(\frac{1-a}{2}\right)^n\right]$, $n \geq 0$, $a = \sqrt{5}$

8. $a_n - a_{n-1} + 2a_{n-2} = 0$, $a_0 = 2$, $a_1 = 7$

Ans. $a_n = 3 \cdot 2^n - (-1)^n$

9. $a_n - 6a_{n-1} + 9a_{n-2} = 0$, $a_0 = 1$, $a_1 = 6$

Ans. $a_n = (1+n)3^n$

10. $y_{n+3} - 6y_{n+2} + 11y_{n+1} - 6y_n = 0$ with initial conditions $y_0 = 2$, $y_1 = 5$, $y_2 = 15$.

Ans. $y_n = 1 - 2^n + 2 \cdot 3^n$

11. $y_{n+3} + 3y_{n+2} + 3y_{n+1} + y_n = 0$ with $y_0 = 1$, $y_1 = -2$ and $y_2 = -1$.

Ans. $y_n = (1 + 3n - 2n^2)(-1)^n$.

12. $y_{n+3} - 2y_{n+2} - y_{n+1} + 2y_n = 0$ with $y_0 = 0$, $y_1 = 1$, $y_2 = 1$. Also find y_{10}.

Ans. $y_n = \frac{1}{3}\{2^n - (-1)^n\}$, $y_{10} = 341$

13. $y_{n+2} - 2y_{n+1} + 2y_n = 0$, with $y_0 = 0$, $y_1 = 1$

Ans. $y_n = \frac{(1+i)^n - (1-i)^n}{2i}$

14. $y_{n+3} - 2y_{n+2} - 5y_{n+1} + 6y_n = 0$

Ans. $y_n = c_1 \cdot 1^n + c_2(-2)^n + c_3 3^n$

15. $y_{n+1} - 2y_n \cos\alpha + y_{n-1} = 0$

Ans. $y_n = c_1 \cos n\alpha + c_2 \sin n\alpha$

16. $a_{m+4} + 16a_m = 0$

Ans. $a_m = 2^m \left[c_1 \cos\frac{m\pi}{4} + c_2 \sin\frac{m\pi}{4} + c_3 \cos\frac{3m\pi}{4} + c_4 \cdot \sin\frac{3m\pi}{4}\right]$

17. $(E^2 + 2E + 4)y_n = 0$

Ans. $y_n = 2^n((c_1 \cos\frac{2n\pi}{3} + c_2 \sin\frac{2n\pi}{3})$

18. Determine D_n as a function of n where

$$D_n = \begin{vmatrix} 2 & 2 & 0 & 0 & \ldots & 0 & 0 \\ 1 & 2 & 2 & 0 & \ldots & 0 & 0 \\ 0 & 1 & 2 & 2 & \ldots & 0 & 0 \\ 0 & 0 & 1 & 2 & \ldots & 0 & 0 \\ \ldots & \ldots & \ldots & \ldots & \ldots & \ldots & \ldots \\ 0 & 0 & 0 & 0 & \ldots & 2 & 2 \\ 0 & 0 & 0 & 0 & \ldots & 1 & 2 \end{vmatrix}$$

Ans. $D_n = 2^{n/2}\left(\cos\frac{n\pi}{4} + \sin\frac{n\pi}{4}\right)$

Hint Solve $D_n - 2D_{n-1} + 2D_{n-2} = 0$.

Non-homogeneous difference equations

EXERCISE

Solve

1. Determine a formula for the sum of the squares of the first n natural numbers.

Ans. $S_n = n(n+1)(2n+1)/6$

Hint: Solve $S_{n+1} - S_n = (n+1)^2$, use $S_1 = 1$.

2. $y_n - y_{n-1} = 3n^2, n \geq 1, y_0 = 7$

Ans. $y_n = 7 + \frac{1}{2}n(n+1)(2n+1)$

3. $y_{n+1} - 3y_n = 357^n, y_0 = 2$

Ans. $y_n = \frac{5}{4}7^{n+1} - \frac{1}{4}3^{n+3}$

4. Tower of Hanoi: $a_{n+1} - 2a_n = 1, a_0 = 0$, find a_{64}.

Ans. $a_n = 2^n - 1$, $a_{64} = 18, 446, 744, 073, 709, 551, 615$

5. A car loan of S rupees is to be paid back in T periods of time. Determine the equi (constant) payment P at the end of each period if r is the interest rate per period for the loan.

Ans. $P = (S \cdot r)/[1 - (1+r)^{-T}]$

Hint: Solve $a_{n+1} - a_n(1+r) = P,\ a_0 = S,\ a_T = 0$ and $0 \leq n \leq T-1$

6. $(E^2 + 2E - 8)a_n = 5n + 3(2^n)$

Ans. $a_n = c_1 2^n + c_2(-4)^n - n - \frac{4}{5} + n2^{n-2}$

7. $a_{n+2} + 2a_{n+1} + a_n = n + 2, a_0 = 0, a_1 = 0$

Ans. $a_n = \frac{1}{4}(3n-1)(-1)^n + \frac{1}{4}(n+1)$

8. $a_{n+2} - 5a_{n+1} + 6a_n = 2n + 1,\ a_0 = 0,\ a_1 = 1$

Ans. $a_n = \frac{5}{2}3^n - 5 \cdot 2^n + n + \frac{5}{2}$

9. $a_{n+2} + 4a_{n+1} - 5a_n = 24n - 8, a_0 = 3, a_1 = -5$

Ans. $a_n = 2n^2 - 4n + 2 + (-5)^n$

10. $a_{n+2} - 4a_{n+1} + 3a_n = 5^n$

Ans. $a_n = c_1 + c_2 \cdot 3^n + \frac{1}{8}5^n$

11. $(E^2 - 4E + 4)a_n = 2^n$

Ans. $a_n = (c_1 + c_2 \cdot n)2^n + n(n-1)2^{n-3}$

12. $(E^2 - 2\cos\alpha E + 1)a_n = \cos\alpha n$

Ans. $a_n = c_1 \cos\alpha n + c_2 \sin\alpha n + n \cdot \frac{\sin(n-1)\alpha}{2\sin\alpha}$

13. $(E^2 - 4)a_n = n^2 + n - 1$

Ans. $a_n = c_1 2^n + c_2(-2)^n - \frac{n^2}{3} - \frac{7n}{9} - \frac{17}{27}$

14. $(E^2 - 2E + 1)a_n = 2^n \cdot n^2$

Ans. $a_n = c_1 + c_2 \cdot n + 2^n(n^2 - 8n + 20)$

15. $a_n - 7a_{n-1} + 10a_{n-2} = 7 \cdot 3^n$ for $n \geq 2$

Ans. $a_n = c_1 2^n + c_2 5^n + \left(\frac{-63}{2}\right) 3^n$

16. $a_n - 6a_{n-1} + 8a_{n-2} = n \cdot 4^n$ with $a_0 = 8$, $a_1 = 22$

Ans. $a_n = 3 \cdot 4^n + 5 \cdot 2^n + n(n-1)4^n$

Simultaneous equations

Solve

17. $x_{n+1} - y_n = 1;\ y_{n+1} - x_n = 1;\ x_0 = 0,\ y_0 = -1$

18. $x_{n+1} + y_n - 3x_n = n, \quad 3x_n + y_{n+1} - 5y_n = 4^n$ with $x_1 = 2, y_1 = 0$

Ans. $x_n = (1.33)2^n - (0.0167)6^n - 0.8n - 0.76 + 4^{n-1}$,
$y_n = (1.33)2^n - (0.05)6^n - 0.6n - 1.36 - 4^{n-1}$

19. $x_{n+1} - 3x_n - 2y_n = -n$,
$-x_n + y_{n+1} - 2y_n = n, x_0 = 0, y_0 = 3$

Ans. $x_n = 2 \cdot 4^n - 2 - \frac{1}{2}n(n-1)$,
$y_n = 4^n + 2 + \frac{1}{2}n(n+1)$

20. $x_{n+1} - y_n - 1 = 0, \quad y_{n+1} + x_n = 0, \quad x_0 = 0, y_0 = -1$.

21.2 Z-TRANSFORMS

Introduction

Z-transform is useful in solving difference equations which represent a discrete system. Z-transform operates on a sequence u_n of discrete integer-valued arguments $n = 0, \pm1, \pm2, \ldots$ unlike the Laplace transform which operates on continuous functions.

Thus Z-transform is the discrete analogue of Laplace transform. Therefore for every operational rule and application of Laplace transform, there corresponds an operational rule and application of Z-transform.

Definition

The Z-transform of a sequence u_n defined for discrete values $n = 0, 1, 2, \cdots$ (and $u_n = 0$ for $n < 0$) is denoted by $Z(u_n)$ and is defined as

$$Z(u_n) = \sum_{n=0}^{\infty} u_n z^{-n} = U(z) \tag{1}$$

where U is a function of z.

Z-transforms exists only when the infinite series in (1) is convergent.

Inverse Z-transform

is denoted by $Z^{-1}(U(z)) = u_n$ determines the sequence u_n which generates the given Z-transform.

Results

Z-transform of some standard sequences

1. $U_n = \{a^n\} = \{1, a, a^2, a^3, \ldots, a^n, \ldots\}$

By definition,

$$Z(u_n) = Z(a^n) = \sum u_n z^{-n} = \sum_{n=0}^{\infty} a^n z^{-n}$$

$$= 1 + \frac{a}{z} + \frac{a}{z^2} + \frac{a}{z^3} + \cdots + \frac{a^n}{z^n} + \cdots = \frac{1}{1 - \frac{a}{z}}$$

$$Z(a^n) = \frac{z}{z-a}.$$

Recurrence formula

2. $u_n = \{n^p\} = \{0, 1^p, 2^p, 3^p, \ldots, n^p, \ldots\}$ where p is a positive integer

By definition

$$Z(n^p) = \sum_{n=0}^{\infty} n^p z^{-n} \tag{i}$$

Replacing p by $p-1$

$$Z(n^{p-1}) = \sum_{n=0}^{\infty} n^{p-1} z^{-n} \tag{ii}$$

Differentiating (ii) w.r.t. z, we get

$$\frac{d}{dz}\left[Z(n^{p-1})\right] = \frac{d}{dz}\left[\sum_{n=0}^{\infty} n^{p-1} z^{-n}\right]$$

$$= \sum_{n=0}^{\infty} n^{p-1} \cdot (-n) z^{-n-1}$$

$$= -z^{-1} \sum_{n=0}^{\infty} n^p \cdot z^{-n}$$

$$= -z^{-1} Z(n^p), \text{ using (i)}$$

Thus $$Z(n^p) = -z \frac{d}{dz}\left[Z(n^{p-1})\right]$$

Special cases of result 1.

3. $Z(1) = \frac{z}{z-1}$

obtained by putting $a = 1$ in result 1.

4. $Z(k) = \frac{kz}{z-1}$ (where k is a constant)

since $z(k) = \sum k z^{-n} = k \frac{1}{1-\frac{1}{z}} = k \frac{z}{z-1}$

5. $Z\left[(-1)^n\right] = \frac{z}{z+1}$

obtained by taking $a = -1$ in result 1.

Special cases of result 2.

6. $Z(n) = \frac{z}{(z-1)^2}$

since with $p = 1$, $Z(n) = -z \cdot \frac{d}{dz}[Z(1)] = -z \frac{d}{dz}\left(\frac{z}{z-1}\right) = -z \cdot \left[\frac{(z-1)\cdot 1 - z \cdot 1}{(z-1)^2}\right] = \frac{z}{(z-1)^2}$

7. $Z(n^2) = \frac{z^2+z}{(z-1)^3}$ by taking $p = 2$ in result 2.

8. $Z(n^3) = \frac{z^3+4z^2+z}{(z-1)^4}$

9. $Z(n^4) = \frac{z^4+11z^3+11z^2+z}{(z-1)^5}$

10. $Z\left(\frac{1}{n!}\right) = \sum_{n=0}^{\infty} \frac{1}{n!} z^{-n} = 1 + \frac{1}{z} + \frac{1}{2!}\frac{1}{z^2} + \frac{1}{3!}\frac{1}{z^3} + \cdots + \frac{1}{n!}\frac{1}{z^n} + \cdots = e^{\left(\frac{1}{z}\right)}$

11. Unit step sequence $u(n) = \begin{cases} 0, & \text{for } n < 0 \\ 1, & \text{for } n \geq 0 \end{cases}$

$$Z(u(n)) = \sum_{n=0}^{\infty} u(n) z^{-n} = \sum_{n=0}^{\infty} 1 \cdot z^{-n}$$

$$= 1 + \frac{1}{z} + \frac{1}{z^2} + \cdots + \frac{1}{z^n} + \cdots$$

$$Z[u(n)] = \frac{1}{1-\frac{1}{z}} = \frac{z}{z-1}$$

12. Unit impulse sequence $\delta(n) = \begin{cases} 1, & \text{for } n = 0 \\ 0, & \text{for } n \neq 0 \end{cases}$

$$Z[\delta(n)] = \sum_{n=0}^{\infty} \delta(n) z^{-n} = 1 + 0 + 0 + 0 \cdots = 1.$$

Properties

I. Linearity

$Z(au_n + bv_n) = aZ(u_n) + bZ(v_n)$

since $\quad Z(au_n + bv_n) = \sum (au_n + bv_n) z^{-n}$

$$= a \sum u_n z^{-n} + b \sum v_n z^{-n}.$$

II. Change of scale (or damping rule)

If $Z(u_n) = U(z)$ then $Z(a^{-n} u_n) = U(az)$
since

$$Z(a^{-n} u_n) = \sum_{n=0}^{\infty} a^{-n} u_n z^{-n} = \sum_{n=0}^{\infty} u_n (az)^{-n} = U(az)$$

Similarly,

$$Z(a^n u_n) = U(z/a)$$

Results from application of damping rule:

13. $Z(na^n) = \frac{az}{(z-a)^2}$

because $Z(n) = \frac{z}{(z-1)^2}$ then $Z(na^n)$ is obtained by replacing z by z/a. So $Z(na^n) = \frac{z/a}{(z/a-1)^2} = \frac{az}{(z-a)^2}$

14. $Z(n^2 a^n) = \frac{az^2 + a^2 z}{(z-a)^3}$

since $Z(n^2) = \frac{z^2+z}{(z-1)^3}$ by damping rule. $Z(n^2 a^n)$ is obtained by replacing z by z/a

i.e., $\quad Z(n^2 a^n) = \dfrac{(z/a)^2 + (z/a)}{[(z/a) - 1]^3} = \dfrac{a(z^2 + az)}{(z-a)^3}$

15. $Z(\cos n\theta) = \frac{z(z - \cos\theta)}{z^2 - 2z\cos\theta + 1}$

Solution:

$$Z(e^{-in\theta}) = Z\left[(e^{-i\theta})^n\right] = Z\left[(e^{-i\theta})^n \cdot 1\right]$$

since $Z(1) = \frac{z}{z-1}$, apply damping rule and replace z by z/a where $a = e^{-i\theta}$ i.e., z by $z/e^{-i\theta}$ or $ze^{i\theta}$. Thus

$$Z\left[(e^{-i\theta})^n \cdot 1\right] = \left.\frac{z}{z-1}\right|_{ze^{i\theta}} = \frac{ze^{i\theta}}{ze^{i\theta} - 1}$$

$$= \frac{z}{z - e^{-i\theta}} = \frac{z(z - e^{i\theta})}{(z - e^{-i\theta})(z - e^{i\theta})}$$

$$= \frac{z[z - \cos\theta - i\sin\theta]}{z^2 - z(e^{i\theta} + e^{-i\theta}) + 1}$$

Thus

$$Z(e^{-in\theta}) = Z(\cos n\theta - i\sin n\theta)$$

$$= \frac{z(z - \cos\theta) - iz\sin\theta}{z^2 - 2z\cos\theta + 1}$$

Since $\qquad \cos\theta = (e^{i\theta} + e^{-i\theta})/2$

Applying linearity property and comparing the real parts on both sides, we get

$$Z(\cos n\theta) = \frac{z(z - \cos\theta)}{z^2 - 2z\cos\theta + 1}$$

Similarly, comparing the imaginary parts

16. $Z(\sin n\theta) = \frac{z\sin\theta}{z^2 - 2z\cos\theta + 1}$

17. $Z(a^n \cos n\theta) = \frac{z(z - a\cos\theta)}{z^2 - 2az\cos\theta + a^2}$ obtained by replacing z by z/a in (15) by damping rule.

Similarly,

18. $Z(a^n \sin n\theta) = \frac{az\sin\theta}{z^2 - 2az\cos\theta + a^2}$

III. Shifting property

a. Shifting u_n to the right

If $Z(u_n) = U(z)$ then $Z(u_{n-k}) = z^{-k} U(z)$ for $k > 0$

This follows from the definition

$$Z(u_{n-k}) = \sum_{n=0}^{\infty} u_{n-k} z^{-n}$$

$$= u_0 z^{-k} + u_1 z^{-(k+1)} + u_2 z^{-(k+2)} + \cdots$$

since $u_n = 0$ for $n < 0$ (so $u_{-k}, u_{1-k}, u_{2-k}, \ldots$ are all 0)

$$= Z^{-k} \sum_{n=0}^{\infty} u_n z^{-n} = z^{-k} U(z)$$

b. Shifting to the left

If $Z(u_n) = U(z)$ then

$$Z(u_{n+k}) = z^k \left[U(z) - u_0 - \frac{u_1}{z} - \frac{u_2}{z^2} \cdots \frac{u_{k-1}}{z^{k-1}}\right]$$

Proof:

$$\begin{aligned}
Z(u_{n+k}) &= \sum_{n=0}^{\infty} u_{n+k} z^{-n} = z^k \sum_{n=0}^{\infty} u_{n+k} z^{-(n+k)} \\
&= z^k \left[u_k z^{-k} + u_{1+k} z^{-(1+k)} + u_{2+k} z^{-(2+k)} + \cdots\right] \\
&= z^k \left[(u_0 + u_1 z^{-1} + u_2 z^{-2} + \cdots + u_{k-1} z^{-(k-1)}) + \right. \\
&\quad \left. + u_k z^{-k} + u_{k+1} z^{-(k+1)+\cdots}\right] - \\
&\quad - z^k \left[u_0 + u_1 z^{-1} + u_2 z^{-2} + \cdots + u_{k-1} z^{-(k-1)}\right]
\end{aligned}$$

$$Z(u_{n+k}) = z^k \left[\sum_{n=0}^{\infty} u_n z^{-n} - \sum_{n=0}^{k-1} u_n z^{-n}\right]$$

$$Z(u_{n+k}) = z^k \left[U(z) - u_0 - u_1 z^{-1} - u_2 z^{-2} \cdots - u_{k-1} z^{-(k-1)}\right]$$

In particular for $k = 1, 2, 3$

19. $Z(u_{n+1}) = z[U(z) - u_0]$

20. $Z(u_{n+2}) = z^2[U(z) - u_0 - u_1 z^{-1}]$

21. $Z(u_{n+3}) = z^3[U(z) - u_0 - u_1 z^{-1} - u_2 z^{-2}]$.

IV. Multiplication by *n*

If $Z(u_n) = U(z)$ then $Z(nu_n) = -z \frac{dU}{dz}(z)$.

Proof:

$$\begin{aligned}
Z(nu_n) &= \sum_{n=0}^{\infty} n \cdot u_n z^{-n} = -z \sum_{n=0}^{\infty} u_n (-n) z^{-n-1} \\
&= -z \sum_{n=0}^{\infty} u_n \frac{d}{dz}(z^{-n}) = -z \sum \frac{d}{dz}(u_n z^{-n}) \\
&= -z \frac{d}{dz}\left(\sum u_n z^{-n}\right) = -z \cdot \frac{d}{dz} U(z)
\end{aligned}$$

In general (by mathematical induction)

$$Z(n^p u_n) = (-z)^p \frac{d^p U(z)}{dz^p}$$

Initial value and final value theorems determine the values of u_n for $n = 0$ and for $n \to \infty$ without the complete knowledge of u_n.

V. Initial value theorem

Theorem: *If $Z(u_n) = U(z)$ then $u_0 = \lim_{z\to\infty} U(z)$*

Proof: By definition

$$U(z) = Z(u_n) = u_0 + \frac{u_1}{z} + \frac{u_2}{z^2} + \cdots + \frac{u_n}{z^n} + \cdots$$

Taking the limit as $z \to \infty$

22. $\lim_{z\to\infty} U(z) = u_0 + 0 + 0 \cdots = u_0$

23. $\lim_{z\to\infty} z[U(z) - u_0] =$

$\lim_{z\to\infty} \left[u_1 + \frac{u_2}{z} + \frac{u_3}{z^2} + \cdots\right] = u_1$

24. $\lim_{z\to\infty} z^2 \left[u(z) - u_0 - \frac{u_1}{z}\right] =$

$\lim_{z\to\infty} \left[u_2 + \frac{u_3}{z} + \frac{u_4}{z^2} + \cdots\right] = u_2.$

VI. Final value theorem

Theorem: *If $Z(u_n) = U(z)$ then*

$$\lim_{n\to\infty} u_n = \lim_{z\to 1} \{(z-1)U(z).\}$$

Proof: Consider

$$\begin{aligned}
Z(u_{n+1} - u_n) &= Z(u_{n+1}) - Z(u_n) \\
&= Z[U(z) - u_0] - U(z) \quad \text{using (19)} \\
&= U(z) \cdot (z-1) - u_0 z
\end{aligned}$$

From definition and the above result, we have

$$(z-1)U(z) - u_0 z = Z(u_{n+1} - u_n) = \sum_{n=0}^{\infty} (u_{n+1} - u_n) z^{-n}$$

As $z \to 1$, we have

$$\begin{aligned}
&\lim_{z\to 1} [(z-1)U(z)] - u_0 \\
&= \lim_{z\to 1} \sum_{n=0}^{\infty} (u_{n+1} - u_n) z^{-n} = \sum_{n=0}^{\infty} (u_{n+1} - u_n) \\
&= \lim_{n\to\infty} [(u_1 - u_0) + (u_2 - u_1) + \cdots + (u_{n+1} - u_n)] \\
&= \lim_{n\to\infty} [u_{n+1} - u_0] = \lim_{n\to\infty} u_{n+1} - u_0
\end{aligned}$$

Thus

$$\lim_{n\to\infty} u_n = \lim_{z\to 1} [(z-1)U(z)].$$

VII. Convolution theorem

Theorem: *If* $Z^{-1}[U(z)] = u_n$ *and* $Z^{-1}[V(z)] = v_n$ *then*

$$Z^{-1}[U(z)\cdot V(z)] = u_n * v_n = \text{convolution of } u_n \text{ and } v_n$$
$$= \sum_{m=0}^{n} u_m v_{n-m}$$

Proof:

$$U(z)\cdot V(z) = \left[\sum_{n=0}^{\infty} u_n z^{-n}\right]\left[\sum_{n=0}^{\infty} v_n z^{-n}\right]$$
$$= \left[u_0 + \frac{u_1}{z} + \frac{u_2}{z^2} + \cdots + \frac{u_n}{z^n} + \cdots\right] \times$$
$$\times \left[v_0 + \frac{v_1}{z} + \frac{v_2}{z^2} + \cdots + \frac{v_n}{z^n} \cdots\right]$$

Collecting the coefficients of z^{-n}

$$= \sum_{n=0}^{\infty} (u_0 v_n + u_1 v_{n-1} + u_2 v_{n-2} + \cdots + u_n v_0) z^{-n}$$
$$= Z(u_0 v_n + u_1 v_{n-1} + \cdots + u_n v_0) \quad \text{by definition}$$
$$= Z\left(\sum_{m=0}^{n} u_m v_{n-m}\right)$$

Taking inverse Z-transform the result follows.

VIII. Region of convergence (R.O.C.)

The region of convergence of Z-transform is the region in the z-plane where the infinite series convergence absolutely.

Thus the region of convergence of a one-sided Z-transform of a right-sided sequence i.e.,

$$U(z) = \sum_{n=0}^{\infty} u_n z^{-n}$$

is $|z| > a$ i.e., the exterior of the circle with centre at origin and of radius a.

Similarly the R.O.C. of

$$U(z) = \sum_{n=-\infty}^{0} u_n z^{-n}$$

is $|z| < a$. Finally the R.O.C. of two-sided Z-transform defined by

$$U(z) = \sum_{n=-\infty}^{\infty} u_n z^{-n}$$

is the annulus region $a < |z| < b$.

IX. Solution of difference equations

Step 1. Take Z-transform on both sides of the given difference equation.

Step 2. Use given conditions and solve for $U(z)$.

Step 3. Apply partial fractions method.

Step 4. Take inverse Z-transform on both sides which results in the given sequence.

Note: A list of standard Z-transforms is presented on page 21.30.

WORKED OUT EXAMPLES

Linearity property

Example 1: Find the Z-transform of $2n + 5\sin\frac{n\pi}{4} - 3a^4$.

Solution: By linearity property

$$Z\left(2n + 5\sin\frac{n\pi}{4} - 3a^4\right)$$
$$= 2Z(n) + 5Z\left(\sin\frac{n\pi}{4}\right) - 3a^4 Z(1)$$
$$= \frac{2\cdot z}{(z-1)^2} + \frac{5\cdot z\cdot \sin\frac{\pi}{4}}{z^2 - 2z\cdot\cos\frac{\pi}{4} + 1} - 3a^4\frac{z}{z-1}$$
$$= \frac{2z}{(z-1)^2} + \frac{5(z/\sqrt{2})}{z^2 - \sqrt{2}z + 1} - 3a^4\frac{z}{z-1}.$$

Example 2: Find $Z(\cos\theta + i\sin\theta)^n$.

Solution: We know that $(\cos\theta + i\sin\theta) = e^{i\theta}$

$$Z(\cos\theta + i\sin\theta)^n = Z\left[(e^{i\theta})^n\right] = \frac{z}{z - e^{i\theta}}$$

since $$Z(a^n) = \frac{z}{z-a}.$$

Example 3: Find $Z\left[(n+1)^2\right]$.

Solution:

$$Z(n+1)^2 = Z(n^2 + 2n + 1) = Z(n^2) + 2Z(n) + Z(1)$$
$$= \frac{z^2+z}{(z-1)^3} + \frac{2\cdot z}{(z-1)^2} + \frac{z}{z-1}$$
$$= \frac{(z^2+z) + 2z(z-1) + z(z-1)^2}{(z-1)^3} = \frac{z^3+z^2}{(z-1)^3}$$

Damping rule

Example 4: Find $Z(e^{-an} \cdot \sin n\theta)$.

Solution: We know that $Z(\sin n\theta) = \frac{z\sin\theta}{z^2-2z\cos\theta+1}$

Now $\quad Z(e^{-an} \cdot \sin n\theta) = Z\left[(e^a)^{-n} \sin n\theta\right]$

Applying damping rule $\quad Z[a^{-n}u_n] = U(az)$

Replace z by az i.e., $e^a z$ then

$$z[e^{-an} \sin n\theta] = \left.\frac{z\sin\theta}{z^2 - 2z\cos\theta + 1}\right|_{z=e^a z}$$
$$= \frac{e^a z \cdot \sin\theta}{e^{2a}z^2 - 2e^a z\cos\theta + 1}.$$

Multiplication by n

Example 5: Find $Z(n\cos n\theta)$.

Solution: We know that $Z(nu_n) = -z\frac{d}{dz}(U(z))$

Here $\quad u_n = \cos n\theta$

and $\quad Z(u_n) = U(z) = \dfrac{z(z-\cos\theta)}{z^2 - 2z\cos\theta + 1}$

so $\quad Z(n\cos n\theta) = -z \cdot \dfrac{d}{dz}\left\{\dfrac{z(z-\cos\theta)}{z^2-2z\cos\theta+1}\right\}$

$$Z(n\cos n\theta) = -z \cdot \frac{[-z^2\cos\theta + 2z - \cos\theta]}{(z^2-2z\cos\theta+1)^2}$$
$$= \frac{z^3\cos\theta - 2z^2 + z\cos\theta}{(z^2-2z\cos\theta+1)^2}.$$

Shift and initial value theorems

Example 6: Find $Z(u_{n+2})$ if $Z(u_n) = \frac{z}{z-1} + \frac{z}{z^2+1}$.

Solution:

Here $\quad Z(u_n) = U(z) = \frac{z}{z-1} + \frac{z}{z^2+1}$

To find u_0, u_1 by initial value theorem

$$u_0 = \lim_{z\to\infty} U(z) = \lim_{z\to\infty}\left[\frac{z}{z-1} + \frac{z}{z^2+1}\right] = 1$$
$$u_1 = \lim_{z\to\infty}\{z\,[U(z) - u_0]\}$$
$$= \lim_{z\to\infty} z \cdot \frac{2z^2 - z + 1}{(z-1)(z^2+1)} = 2$$

From shift to left we know that

$$Z(u_{n+2}) = z^2\left[U(z) - u_0 - u_1 z^{-1}\right]$$
$$Z(u_{n+2}) = z^2\left[\frac{z}{z-1} + \frac{z}{z^2+1} - 1 - \frac{2}{z}\right]$$
$$= z^3\left[\frac{1}{z-1} + \frac{1}{z^2+1} - \frac{(z+2)}{z^2}\right]$$
$$Z(u_{n+2}) = \frac{z(z^2-z+2)}{(z-1)(z^2+1)}.$$

Inverse transform by partial fractions

Example 7: Find the inverse Z-transform of $\frac{4z^2-2z}{z^3-5z^2+8z-4}$.

Solution: Consider

$$\frac{4z^2-2z}{z^3-5z^2+8z-4} = \frac{2z(2z-1)}{(z-1)(z-2)^2}$$

By partial fractions

$$\frac{2z-1}{(z-1)(z-2)^2} = \frac{A}{z-1} + \frac{B}{z-2} + \frac{C}{(z-2)^2}$$
$$(2z-1) = A(z-2)^2 + B(z-1)(z-2) + C(z-1)$$

when $z = 2, 3 = C$; when $z = 1$, $A = 1$ and when $z = 0$

$$-1 = 4A + 2B - C \qquad \therefore \quad B = -1$$

Thus

$$Z^{-1}\left[\frac{4z^2-2z}{(z-1)(z-2)^2}\right]$$
$$= Z^{-1}\left[2z \cdot \frac{1}{z-1}\right] + Z^{-1}\left[2z\frac{-1}{z-2}\right]$$
$$+ Z^{-1}\left[2z \cdot \frac{3}{(z-2)^2}\right]$$
$$= 2 \cdot 1 - 2 \cdot 2^n + 3z^{-1}\left[\frac{2z}{(z-2)^2}\right]$$
$$= 2 - 2^{n+1} + 3n \cdot 2^n$$

since $Z(na^n) = \dfrac{az}{(z-a)^2}$

Region of convergence

Example 8: Find $Z^{-1}\left\{(z-5)^{-3}\right\}$ when $|z|>5$. Determine the region of convergence.

Solution: Consider

$$(z-5)^{-3}=\frac{1}{(z-5)^3}=\frac{1}{z^3}\frac{1}{\left(1-\frac{5}{z}\right)^3}$$

Expanding by Binomial series which is valid when $|\frac{5}{z}|<1$ or $|z|>5$, we have

$$\frac{1}{(z-5)^3}=\frac{1}{z^3}\left[1+3\left(\frac{5}{z}\right)+\frac{3\cdot 4}{1\cdot 2}\cdot\left(\frac{5}{z}\right)^2+\frac{3\cdot 4\cdot 5}{1\cdot 2\cdot 3}\cdot\left(\frac{5}{z}\right)^3+\frac{3\cdot 4\cdot 5\cdot 6}{1\cdot 2\cdot 3\cdot 4}\left(\frac{5}{z}\right)^4+\cdots\right]$$

$$\frac{1}{(z-5)^3}=\frac{1}{2}\cdot\frac{1}{z^3}\left[1\cdot 2\left(\frac{5}{z}\right)^0+2\cdot 3\left(\frac{5}{z}\right)^1+3\cdot 4\left(\frac{5}{z}\right)^2+4\cdot 5\left(\frac{5}{z}\right)^3+5\cdot 6\left(\frac{5}{z}\right)^4+\cdots\right]$$

$$=\frac{1}{2}\sum_{m=0}^{\infty}(m+1)(m+2)5^m\cdot z^{-m-3};\ \text{put } m+3=n$$

$$U(z)=(z-5)^{-3}=\frac{1}{2}\sum_{n=3}^{\infty}(n-1)(n-2)5^{n-3}z^{-n}$$

$$=\sum_{n=0}^{\infty}u_n z^{-n}$$

Taking inverse Z-transform

$$Z^{-1}[U(z)]=Z^{-1}\left[(z-5)^{-3}\right]=u_n$$

$$=\frac{1}{2}(n-1)(n-2)5^{n-3}\quad\text{for } n\geq 3$$

$$=0\quad\text{for } n<3$$

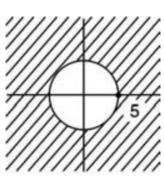

Fig. 21.1

The region of convergence is the exterior of the circle $|z|=5$ i.e., with centre at origin and of radius 5 (Fig. 21.1).

Convolution

Example 9: Using convolution theorem, find the inverse Z-transform of $\left(\frac{z}{z-a}\right)^3$. Deduce for $\left(\frac{z}{z-1}\right)^3$.

Solution: We know that

$$Z^{-1}\left\{\frac{z}{z-a}\right\}=a^n,\ Z^{-1}\left\{\frac{z}{z-a}\right\}=a^n$$

So

$$Z^{-1}\left\{\frac{z^2}{(z-a)^2}\right\}$$

$$=Z^{-1}\left\{\frac{z}{z-a}\cdot\frac{z}{z-a}\right\}=a^n*a^n$$

by convolution theorem

$$=\sum_{m=0}^{n}a^m a^{n-m}=a^n\sum a^m\cdot a^{-m}=a^n\sum_{m=0}^{n}1$$

$$=a^n\cdot[1+1+1+\cdots+1]=a^n(n+1).$$

Applying convolution theorem again

$$Z^{-1}\left\{\frac{z^3}{(z-a)^3}\right\}=Z^{-1}\left\{\frac{z^2}{(z-a)^2}\cdot\frac{z}{z-a}\right\}$$

$$=\left[a^n\cdot(n+1)\right]*a^n$$

$$=\sum_{m=0}^{n}a^m\cdot(m+1)\cdot a^{n-m}$$

$$=a^n\sum_{m=0}^{n}(m+1)$$

$$Z^{-1}\left[\left(\frac{z}{z-a}\right)^3\right]=a^n\cdot[1+2+3+\cdots+(n+1)]$$

$$=a^n\cdot\frac{1}{2}\cdot(n+1)(n+2)$$

Put $a=1$, then

$$Z^{-1}\left\{\frac{z}{(z-1)^3}\right\}=\frac{1}{2}(n+1)(n+2).$$

Solution of difference equations

Example 10: Using Z-transform, solve the difference equation $u_{n+2}-4u_{n+1}+3u_n=5^n$.

Solution: Take Z-transform on both sides of the difference equation

$$Z(u_{n+2}) - 4Z(u_{n+1}) + 3Z(u_n) = Z(5^n)$$

Denote $Z(u_n) = U(z)$, then

$$Z(u_{n+2}) = z^2\left[U(z) - u_0 - u_1 z^{-1}\right]$$
$$Z(u_{n+1}) = z[U(z) - u_0]$$

Substituting these values

$$z^2\left[U(z) - u_0 - u_1 z^{-1}\right] - 4z\,[U(z) - u_0] + 3U(z) = \frac{z}{z-5}$$
$$U(z)\left[z^2 - 4z + 3\right] - u_0\left[z^2 - 4z\right] - u_1 z = \frac{z}{z-5}$$

Solving

$$\frac{U(z)}{z} = \frac{1}{(z-5)(z-1)(z-3)} + \frac{u_0(z-4) + u_1}{(z-1)(z-3)}$$

By partial fractions

$$\frac{U(z)}{z} = \left[\frac{A}{(z-1)} + \frac{B}{(z-3)} + \frac{C}{(z-5)}\right] + \left[\frac{D}{(z-1)} + \frac{E}{z-3}\right]$$

$$U(z) = \left[\frac{1}{8}\frac{z}{z-1} - \frac{1}{4}\frac{z}{z-3} + \frac{1}{8}\frac{z}{z-5}\right] + \left[\frac{(u_1 - 3u_0)z}{2(z-1)} + \frac{3z(u_0 - u_1)}{2(z-3)}\right]$$

Taking inverse Z-transform on both sides

$$u_n = Z^{-1}[U(z)] = \frac{1}{8}\cdot 1 - \frac{1}{4}3^n + \frac{1}{8}5^n + \left(\frac{u_1 - 3u_0}{2}\right)1 + \frac{3(u_0 - u_1)}{2}3^n$$

$$u_n = C_1 + C_2 3^n + \frac{1}{8}5^n \quad \text{is the solution}$$

Here $\quad C_1 = \frac{1}{8} + \frac{u_1 - 3u_0}{2},\ C_2 = \frac{3(u_0 - u_1)}{2} - \frac{1}{4}.$

Example 11: Solve $u_{n+2} + 2u_{n+1} + u_n = n$ with $u_0 = u_1 = 0$

Solution: Taking Z-transform on both sides

$$Z(u_{n+2}) + 2Z(u_{n+1}) + Z(u_n) = Z(n)$$

$$z^2\left[U(z) - u_0 - u_1 z^{-1}\right] + 2z[U(z) - u_0] + U(z) = \frac{z}{(z-1)^2}$$

Putting $u_0 = u_1 = 0$ and solving

$$U(z) = \frac{z}{(z-1)^2(z+1)^2}$$

By partial fractions

$$\frac{1}{(z-1)^2(z+1)^2} = \frac{A}{(z-1)} + \frac{B}{(z-1)^2} + \frac{C}{(z+1)} + \frac{D}{(z+1)^2}$$

we get $A = -\frac{1}{4}$, $B = C = D = \frac{1}{4}$ so

$$U(z) = \frac{1}{4}\left[-\frac{z}{z-1} + \frac{z}{(z-1)^2} + \frac{z}{z+1} + \frac{z}{(z+1)^2}\right]$$

Taking inverse Z-transform on either side

$$u_n = \frac{1}{4}\left[-1^n + n + (-1)^n - n(-1)^n\right]$$
$$u_n = \left(\frac{n-1}{4}\right)\left[1 - (-1)^n\right].$$

Exercise

Find the Z-transform:

1. $\frac{1}{(n+2)!}$ *Ans.* $z^2(e^{\frac{1}{z}} - 1 - z^{-1})$
2. $\cosh n\theta$ *Ans.* $\frac{z(z-\cosh\theta)}{z^2-2z\cosh\theta+1}$
3. $a^n \cdot \sinh n\theta$

   Hint: Use damping rule for $z(\sinh n\theta) = \frac{z\sinh\theta}{z^2-2z\cosh\theta+1}$

Ans. $\frac{az\cdot\sinh\theta}{z^2-2az\cdot\cosh\theta+a^2}$

4. $a^n e^{-a}/n!$ *Ans.* $e^{a(z^{-1}-1)}$
5. $\sin(n+1)\theta$ *Ans.* $\frac{z^2\sin\theta}{z^2-2z\cos\theta+1}$
6. $\frac{1}{n+1}$ *Ans.* $z\ln\left(\frac{z}{z+1}\right)$
7. Find u_2, u_3 if $U(z) = (2z^2+5z+14)/(z-1)^4$.

Ans. $u_0 = 0, u_1 = 0, u_2 = 2, u_3 = 13$

8. Find u_2, u_3 when $U(z) = (5z^2 + 3z + 12)/(z-1)^4$.

Ans. $u_0 = 0, u_1 = 0, u_2 = 5, u_3 = 23$

9. Determine u_2 where $U(z) = (2z^2 + 3z + 4)/(z-3)^3$

Ans. $u_0 = 0, u_1 = 2, u_2 = 21$

10. Use convolution theorem to find the inverse Z-transform of $z^2/[(z-a)(z-b)]$.

Ans. $(a^{n+1} - b^{n+1})/(a-b)$

11. By convolution, evaluate $Z^{-1}[z^2/(z-a)^2]$

Ans. $(n+1)a^n$

Find the inverse Z-transform of

12. $\frac{2z^2+3z}{(z+2)(z-4)}$

Ans. $\frac{1}{6}(-2)^n + \frac{11}{6}4^n$

13. $\frac{z^3-20z}{(z-2)^3(z-4)}$

Ans. $\frac{1}{2}(2^n + 2\cdot n^2 2^n) - 4^n$

14. $\frac{z}{z^2+11z+24}$

Ans. $\frac{1}{5}\left[(-3)^n - (-8)^n\right]$

15. $\frac{z}{z^3-z^2+z-1}$

Ans. $\frac{1}{2}\left[1 - \cos\left(\frac{n\pi}{2}\right) - \sin\left(\frac{n\pi}{2}\right)\right]$

16. $\frac{z}{(z+3)^2(z-2)}$

Ans. $-\frac{1}{25}(-3)^n - \frac{1}{5}n(-3)^n + \frac{1}{25}2^n$

17. $\frac{2(z^2-5z+6.5)}{(z-2)(z-3)^2}$ for $2 < |z| < 3$

Ans. $u_n = 2^{n-1}$, for $n \geq 1$

$u_n = -(n+2)3^{n-2}$, for $n \leq 0$

18. $\frac{1}{(z-2)(z-3)}$ for
   i. $|z| < 2$
   ii. $2 < |z| < 3$
   iii. $|z| > 3$

Ans. **i.** $-\left(\frac{1}{3} + \frac{z}{3^2} + \frac{z^2}{3^3} + \frac{z^3}{3^4} + \cdots\right.$
$\left. + \left(\frac{1}{2} + \frac{z}{2^2} + \frac{z^2}{2^3} + \frac{z^3}{2^4} + \cdots\right)\right)$

ii. (-2^{n-1}) for $n > 0$

iii. $3^{n-1} - 2^{n-1}$, $n \geq 1$, 0 for $n < 0$

Using Z-transform solve the difference equation:

19. $u_{n+2} + 4u_{n+1} + 3u_n = 3^n$ with $u_0 = 0, u_1 = 1$

Ans. $u_n = \frac{3}{8}(-1)^n + \frac{1}{24}3^n - \frac{5}{12}(-3)^n$

20. $u_{n+2} + 6u_{n+1} + 9u_n = 2^n$ with $u_0 = u_1 = 0$

Ans. $u_n = \frac{1}{25}\left[2^n - (-3)^n + \frac{5}{3}n(-3)^n\right]$

21. $u_{n+2} - 5u_{n+1} + 6u_n = y_n$ with $u_0 = 0, u_1 = 1$ where $y(n) = 1$ for $n = 0, 1, 2, 3, \ldots$

Ans. $u_n = \frac{1}{2} - 2(2)^n + \frac{3}{2}(3)^n$

22. $u_n + \frac{1}{4}u_{n-1} = y_n + \frac{1}{3}y_{n-1}$ where y_n is a unit step sequence.

Ans. $\frac{1}{12}\left(-\frac{1}{4}\right)^{n-1}$

23. $u_{n+2} - 2u_{n+1} + u_n = 3n + 5$

Ans. $\frac{1}{2}n(n-1)(n+3) + C_0 + C_1 n$ where $C_0 = u_0, C_1 = u_1 - u_0$

24. $4u_n - u_{n+2} = 0$ with $u_0 = 0, u_1 = 2$

Ans. $u_n = (-2)^{n-1} + 2^{n-1}$

25. $u_{n+2} - 2u_{n+1} + u_n = 2^n$ with $u_0 = 2, u_1 = 1$

Ans. $u_n = 2^n + 1 - 2n$.

21.3 STANDARD Z-TRANSFORMS

	Sequence u_n (for $n \geq 0$)	$U(z)$: Z-transform
1.	n	$z/(z-1)^2$
2.	n^2	$(z^2+z)/(z-1)^3$
3.	n^p	$-z\frac{d}{dz}\left[Z(n^{p-1})\right]$
	(p is positive integer)	
4.	a^n	$z/(z-a)$
5.	nu_n	$-z\frac{d}{dz}[U(z)]$
6.	$a^n u_n$	$U(z/a)$
7.	u_{n+1}	$z[U(z) - u_0]$
8.	u_{n+2}	$z^2\left[U(z) - u_0 - u_1 z^{-1}\right]$
9.	u_{n+3}	$z^2\left[U(z) - u_0 - u_1 z^{-1} - u_2 z^{-2}\right]$
10.	u_{n-k}	$z^{-k}U(z)$
11.	u_0	$\lim_{z\to\infty} U(z)$
12.	$\lim_{n\to\infty} u_n$	$\lim_{z\to 1}[(z-1)U(z)]$
13.	$\sin n\theta$	$\frac{z\sin\theta}{z^2-2z\cos\theta+1}$
14.	$\cos n\theta$	$\frac{z(z-\cos\theta)}{z^2-2z\cos\theta+1}$